



Let's Build, Summer 2023!

By: Derek Hissong, PE Intern





A Bit About Me

- USD 3rd Year Mechanical Engineering student-athlete
- Hometown: Seattle, WA



The USD Wellness Center & NCAA Practice Center

- 80,000 sqft
- \$90,000,000
- 18th major project for USD
- Tucker Sadler & Populous
- My favorite feature(s): sand volleyball court, learning kitchen



The Day-to-Day Action

Activities I've Done:

- RFIs
- Submittals
 - Workflows
 - Review meetings
- VPlanner
- T&M Ticket Review
- Traffic Flagger certified



RFI #0164: Fan Coil Unit Quantity and Designation Confirmation

[General](#) [Related Items \(3\)](#) [Emails \(2\)](#) [Change History](#)

REQUEST

Subject

Fan Coil Unit Quantity and Designation Confirmation

Question

Reference: M201.1A, M201.1B, M202.1A, M202.1B, M203.1A, M203.1B, M805

Floorplans on M201.1A, M201.1B, M202.1A, M202.1B, M203.1A, M203.1B do not have matching fan coil unit count or designation as shown on mechanical schedule M805.

Mechanical floorplans show quantity of (10) fan coil units with designations:

Floor 1: FC B1-3, FC B1-B, FC B1.1, FC B1-3; Floor 2: FC 2-11, FC 2-12, FC 1-23, FC 2-6; Floor 3: FC 1-24, FC 3-1B

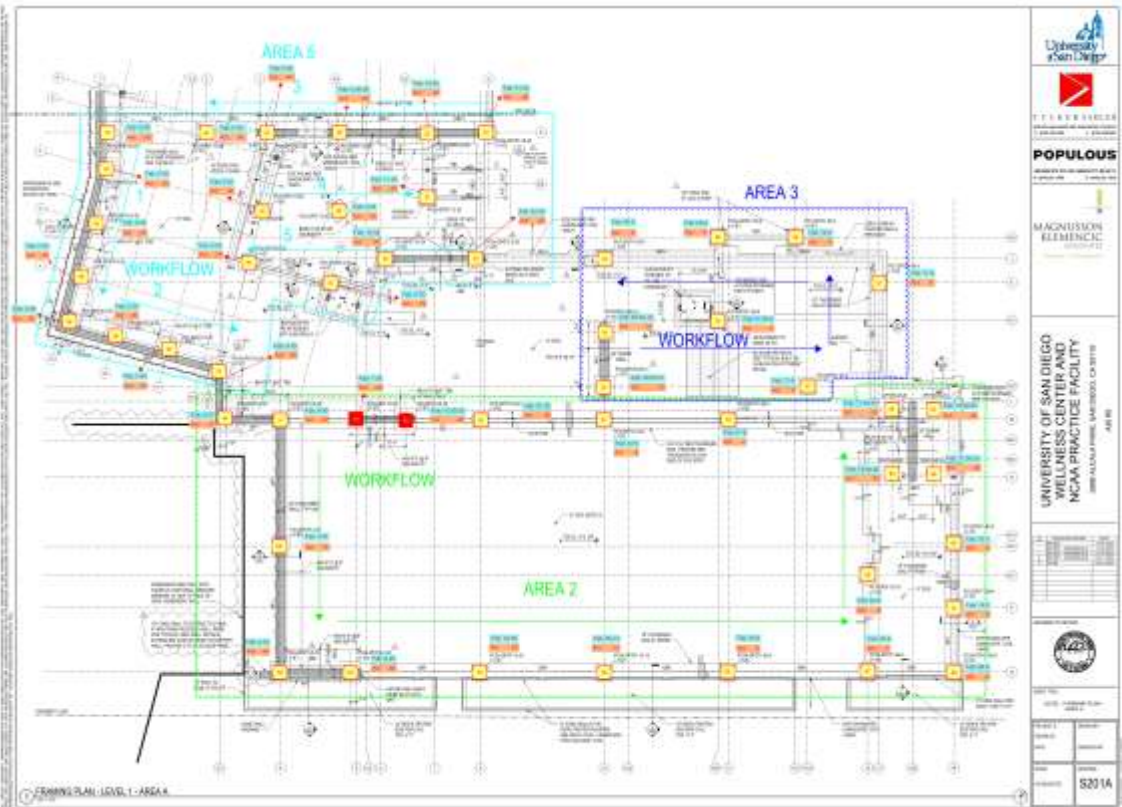
Mechanical schedule (M805) shows (9) fan coil units with different designations:

Floor 1: FC B1-1, FC B1-2; Floor 2: FC A2-1, FC A2-2, FC A2-3, FC A2-4; Floor 3: FC A3-1, FC A3-2, FC B3-1

1. Please provide updated mechanical floorplans and/or mechanical schedule M805 showing matching quantities and designations of all fan coil units.

Attachments

[RFI-0164 - Fan Coil Unit Quantity and Designation Confirmation.pdf](#)



CJA Workflow Plan



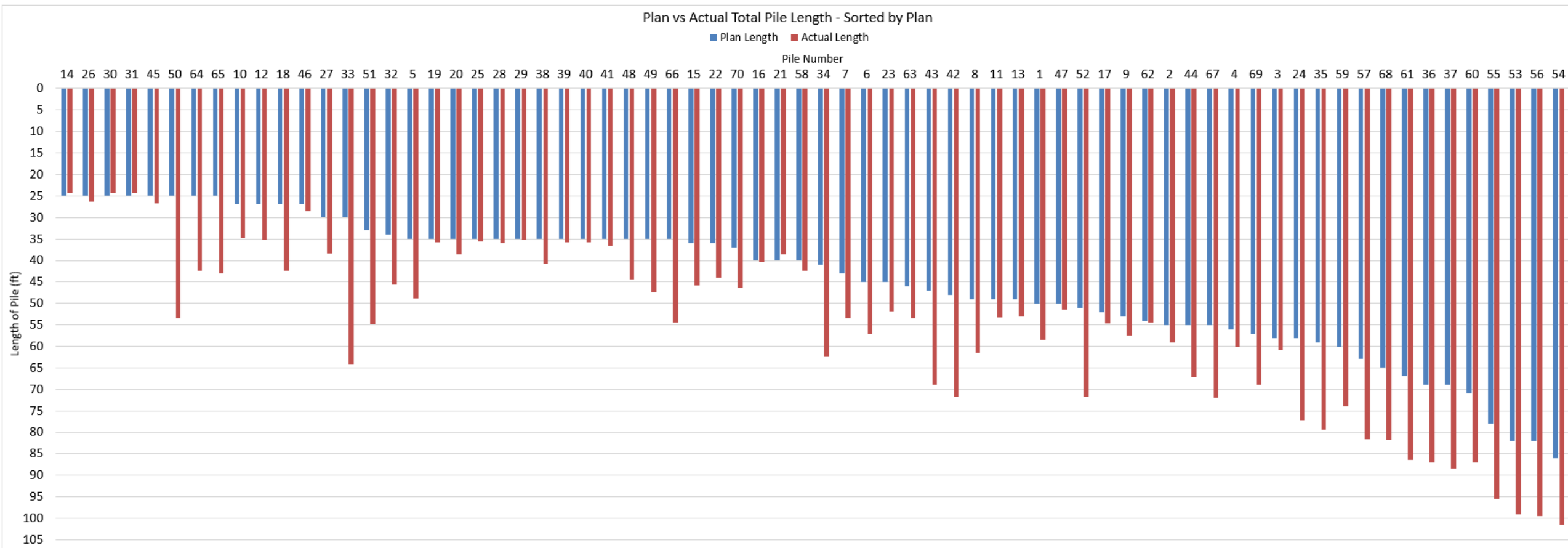
Condon Johnson Piling

Pile Rebar Cage Extensions

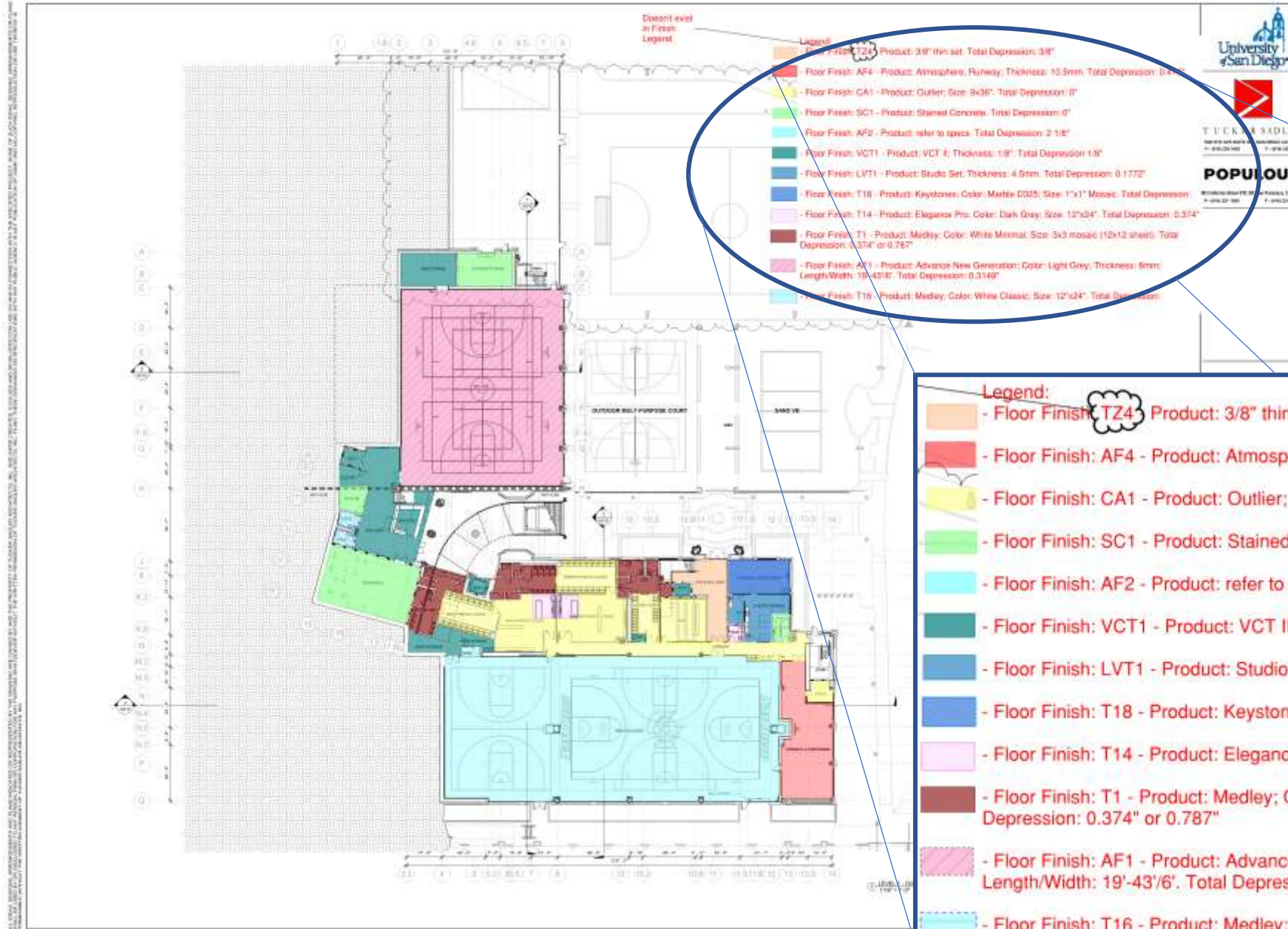
Piles Drilled: 70

Embedment Depth Computations and Comparisons				Depth to Embedment Computations and Comparisons				Total Pile Length Computations and Comparisons					
Pile Number	Actual Depth	Plan Depth	Difference	Pile Number	Actual	Plan	Difference	Pile Designation	Pile Number	Plan Leng	Actual Leng	Differen	Price Add (\$)
1	28.00	9.00	19.00	1	30.50	41.00	-10.50	DP3T1	1	50.00	58.50	8.50	2618.00
2	19.61	9.00	10.61	2	39.50	46.00	-6.50	DP3T1	2	59.00	59.11	4.11	1265.88
3	21.50	9.00	12.50	3	39.50	49.00	-9.50	DP3T1	3	58.00	61.00	3.00	924.00
4	19.00	9.00	10.00	4	41.04	47.00	-5.96	DP3T1	4	56.00	60.04	4.04	1244.32
5	18.00	9.00	9.00	5	30.75	28.00	4.75	DP3T2	5	35.00	48.75	13.75	4427.50
6	25.00	15.00	10.00	6	32.00	30.00	2.00	DP3T3	6	45.00	57.00	12.00	4440.00
7	22.50	9.00	13.50	7	31.00	34.00	-3.00	DP3T1	7	43.00	53.50	10.50	3234.00
8	15.00	9.00	6.00	8	46.42	40.00	6.42	DP3T1	8	49.00	61.42	12.42	3825.36
9	14.40	9.00	5.40	9	43.10	44.00	-0.90	DP3T1	9	53.00	57.50	4.50	1386.00
10	11.00	9.00	2.00	10	23.78	18.00	5.78	DP3T1	10	27.00	34.78	7.78	2396.24
11	15.00	9.00	6.00	11	38.22	40.00	-1.78	DP3T1	11	49.00	53.22	4.22	1299.76
12	12.00	12.00	0.00	12	23.18	15.00	8.18	DP3T1	12	27.00	35.18	8.18	2519.44
13	10.00	9.00	1.00	13	43.08	40.00	3.08	DP3T1	13	49.00	53.08	4.08	1256.64
14	25.00	25.00	0.00	14	-0.57	0.00	-0.57	DP3T1	14	25.00	24.33	-0.67	-206.36
15	11.90	9.00	2.90	15	33.99	27.00	6.99	DP3T1	15	36.00	45.89	9.89	3046.12
16	40.50	40.00	0.50	16	-0.14	0.00	-0.14	DP4T4	16	40.00	40.36	0.36	220.68
17	54.10	52.00	2.10	17	0.55	0.00	0.55	DP4T3	17	52.00	54.65	2.65	1566.15
18	10.00	9.00	1.00	18	32.39	18.00	14.39	DP3T1	18	27.00	42.39	15.39	4740.12
19	34.74	35.00	-0.26	19	1.00	0.00	1.00	DP4T1	19	35.00	35.74	0.74	375.18
20	38.60	35.00	3.60	20	0.08	0.00	0.08	DP4T1	20	35.00	38.68	3.68	1865.76
21	38.60	40.00	-1.40	21	-0.06	0.00	-0.06	DP4T4	21	40.00	38.54	-1.46	-894.98
22	38.00	36.00	2.00	22	5.98	0.00	5.98	DP4T1	22	36.00	43.96	7.96	4045.86
23	41.00	40.00	1.00	23	10.78	5.00	5.78	DP4T4	23	45.00	51.78	6.78	4156.14
24	43.00	40.00	3.00	24	35.19	18.00	17.19	DP4T3	24	58.00	77.19	19.19	11341.29
25	34.58	35.00	-0.42	25	1.00	0.00	1.00	DP4T3	25	35.00	35.58	0.58	342.78
26	27.00	25.00	2.00	26	-0.60	0.00	-0.60	DP3T1	26	25.00	26.40	1.40	431.20
27	27.50	25.00	2.50	27	10.67	5.00	5.67	DP3T1	27	30.00	38.37	8.37	2577.96
28	34.90	35.00	-0.10	28	1.00	0.00	1.00	DP4T2	28	35.00	35.90	0.90	495.00
29	34.19	35.00	-0.81	29	1.00	0.00	1.00	DP4T3	29	35.00	35.19	0.19	112.29
30	25.00	25.00	0.00	30	-0.62	0.00	-0.62	DP3T3	30	25.00	24.38	-0.62	-229.40

Being Proactive as the GC!



Flooring Take-Off

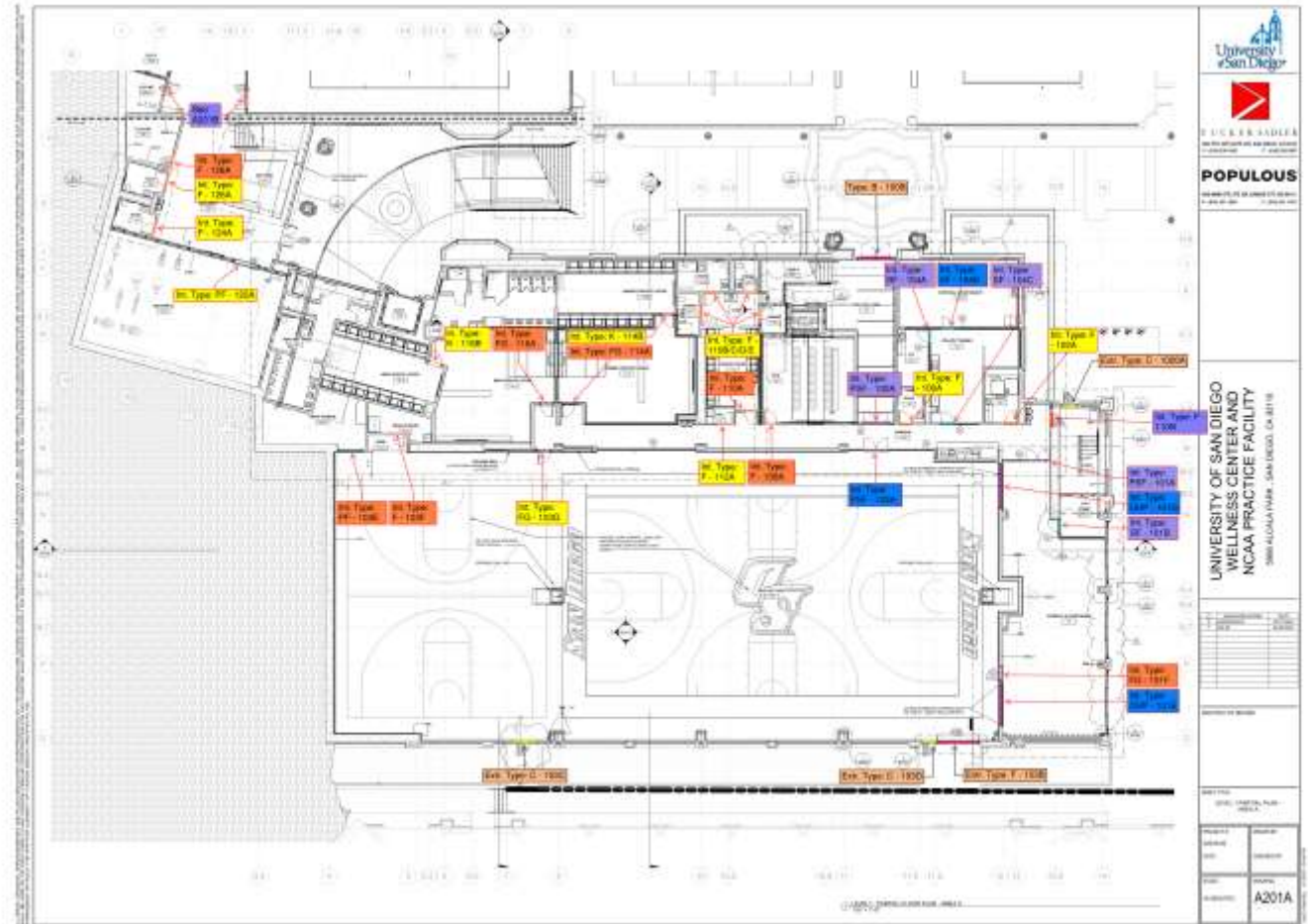


Legend:	
	- Floor Finish: TZ4 - Product: 3/8" thin set. Total Depression: 3/8"
	- Floor Finish: AF4 - Product: Atmosphere, Runway; Thickness: 10.5mm. Total Depression: 0.413"
	- Floor Finish: CA1 - Product: Outlier; Size: 9x36". Total Depression: 0"
	- Floor Finish: SC1 - Product: Stained Concrete. Total Depression: 0"
	- Floor Finish: AF2 - Product: refer to specs. Total Depression: 2 1/8"
	- Floor Finish: VCT1 - Product: VCT II; Thickness: 1/8". Total Depression 1/8"
	- Floor Finish: LVT1 - Product: Studio Set; Thickness: 4.5mm. Total Depression: 0.1772"
	- Floor Finish: T18 - Product: Keystones; Color: Marble D325; Size: 1"x1" Mosaic. Total Depression:
	- Floor Finish: T14 - Product: Elegance Pro; Color: Dark Grey; Size: 12"x24". Total Depression: 0.374"
	- Floor Finish: T1 - Product: Medley; Color: White Minimal; Size: 3x3 mosaic (12x12 sheet). Total Depression: 0.374" or 0.787"
	- Floor Finish: AF1 - Product: Advance New Generation; Color: Light Grey; Thickness: 8mm; Length/Width: 19'-43'/6". Total Depression: 0.3149"
	- Floor Finish: T16 - Product: Medley; Color: White Classic; Size: 12"x24". Total Depression:

Door Take-Off

Interior Door Sub Color Legend

Design	Helix	Centex	Woods	Design + Helix	Centex + Helix	Woods + Helix
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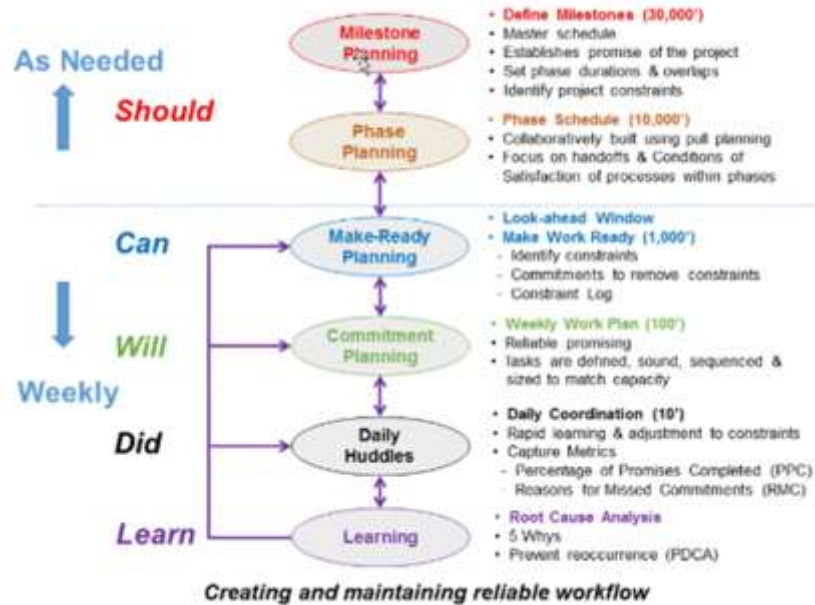


Major Learning Points

- Started at certified pad in terms of knowledge
- Dynamic of each position within team
- Professional vs personal conversation
- Clearing constraints and running free!



Internship Enhancing Experiences



PSSI Yard Shotcrete



SD Intern Site Tour with Paul
Sites Visited: 4

Intern Days

- Right: Joining the 1% Club!
- Middle: 2nd Place! HQ Bootcamp



Plus/Deltas

Plus

- Intern bonding/HQ days
- Openness to questions by all of R&S
- Explaining the WHY along with the HOW
- Meaningful work
- Hosting the Safety Dinner

Deltas

- Unsure how to inquire about other R&S fields

My Takeaways



R&S July Safety
Dinner Kahoot Win

Company Values

- Client focused
- Safety
- Technology Driven



Other R&S Projects for USD

Technology to Enhance
the Project

